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Research Proposal for the Ph.D. Degree

Heart-Brain Modulation Across Sleep Stages and Clinical Diagnoses in a Large Cohort

מודולציה בין הלב למוח בשלבי שינה במגוון אבחנות קליניות במדגם רחב

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Submission date: _____

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Summary

The heartbeat-evoked potential (HEP) is an R-peak-locked EEG deflection that indexes cortical processing of cardiac afferent signals and is used as a neural marker of interoception. Prior work reports that HEP magnitude follows a vigilance-state gradient across sleep and rises with age, but these effects have been measured in small, single-site samples, have not been topographically mapped, and have never been compared against clinical diagnosis in the same framework. **Objectives.** This project asks (i) whether the sleep-stage and age HEP effects share a cortical generator and topography, and (ii) whether patients with cardiovascular, cerebrovascular or neurodegenerative diagnoses show an HEP sleep-stage profile that departs from the diagnosis-free reference, or merely a rescaled version of it. **Methodology.** Using the Human Sleep Project, a multi-centre polysomnography corpus of ~90,000 patients, HEP epochs are extracted per sleep stage from a standardised EEG/ECG cleaning pipeline. Sleep-stage and age contrasts are tested with nonparametric cluster-mass permutation tests, electrode by electrode. Diagnosis groups are then compared to a demographically matched diagnosis-free cohort using the same statistics plus a waveform-alignment permutation test and a cross-validated classifier. **Preliminary results.** In 2,136 patients the REM–deep-sleep HEP difference is robust ($p = 0.005$, 15/19 electrodes) and an older-vs-younger REM contrast matches it in magnitude ($p = 0.005$, 9/19 electrodes). In a first diagnosis analysis the sleep-stage HEP delta of diagnosed subgroups is nearly collinear with the diagnosis-free delta (Pearson $r > 0.999$), suggesting diagnosis rescales rather than reshapes the interoceptive gradient. The proposal develops this into a full topographic and clinical characterisation.

1. Introduction

1.1 The heartbeat-evoked potential and cortical interoception

Every cardiac cycle generates afferent volleys, from arterial baroreceptors, mechanoreceptors and chemoreceptors, that reach the nucleus of the solitary tract and, via thalamic and parabrachial relays, the insular, somatosensory and cingulate cortices (Critchley and Garfinkel, 2017). Time-locking the EEG to the electrocardiographic R-peak and averaging isolates the heartbeat-evoked potential (HEP), a low-amplitude deflection typically largest 200–400 ms after the R-peak over fronto-central and central electrodes (Park and Blanke, 2019). HEP amplitude scales with attention directed to the heartbeat, with interoceptive accuracy, and with self-referential processing in the default-mode network (Babo-Rebelo et al., 2016), and it is therefore widely used as a non-invasive index of how strongly the cortex represents the internal state of the body.

The principal methodological threat to the HEP is the cardiac field artifact (CFA): the electrical field of the heart volume-conducts to the scalp with a fixed phase relationship to the same R-peak used for epoching, so it survives trial averaging exactly as a genuine cortical response does (Park and Blanke, 2019). Current methodological guidance is to exclude a short window around the QRS complex, commonly ± 50 ms, from every statistical test, and to report the retained interval, the number of heartbeats, and channel-level pre/post-cleaning metrics (Steinfath et al., 2026). Any HEP study, and in particular any group comparison, has to demonstrate that a reported effect peaks outside this excluded window and is not tracking a between-group difference in cardiac electrophysiology or thoracic geometry.

1.2 The HEP changes with vigilance state and with age

Central–autonomic coupling is not static across the night. Slow oscillations and sleep spindles gate autonomic outflow, heart-rate variability shifts toward vagal dominance in non-REM sleep, and phasic REM is accompanied by autonomic surges (de Zambotti et al., 2018). Against this background, Lechinger et al. (2015) reported that heartbeat-related EEG amplitude follows a vigilance gradient, with larger responses in wakefulness and REM than in deep sleep, and interactions with spindles and slow oscillations. Independently, ageing has been linked to a larger HEP: older adults show higher HEP amplitude than young adults during rest and during attention orienting (Kamp et al., 2021; Aprile et al., 2025), a change that has been interpreted as altered interoceptive gain or reduced afferent inhibition.

Three gaps follow directly from this literature. First, the sleep-stage and age effects have been established in separate small cohorts (tens of participants), so it is unknown whether they reflect one modulatory axis, arousability, or two. Second, neither effect has been topographically mapped with adequate electrode coverage and multiple-comparison

control, so it is unclear whether they share a cortical generator. Third, and most important for clinical translation, no study has asked whether these normative gradients are preserved, amplified or distorted in patients with disease. This matters because the diseases most likely to affect the HEP, atrial fibrillation, heart failure, stroke and neurodegeneration, are exactly the conditions that also alter sleep architecture and autonomic tone, so a naive case–control HEP comparison that is not matched for age and not stratified by sleep stage can easily mistake a physiological shift for a disease signature.

1.3 Why a large clinical polysomnography corpus

Answering these questions requires a sample that is simultaneously large (for electrode-wise permutation statistics and demographic matching), diagnostically annotated, and recorded with synchronous EEG and ECG across sleep stages. Clinical polysomnography corpora meet all three requirements. The Human Sleep Project provides ~90,000 overnight recordings, of which a subset (~2,443) carry the full 19-electrode 10–20 montage needed for topographic analysis, together with linked diagnosis codes. Multi-hospital clinical corpora have already been used successfully for population-scale neurophysiology in epilepsy imaging (Cafri et al., 2025), and the same design is appropriate here: it trades the homogeneity of a laboratory sample for the statistical power and diagnostic range that the present questions need, provided that cohort heterogeneity is handled by explicit matching and by re-using one fixed, quality-controlled analysis pipeline for every contrast.

Problem statement. The HEP is being adopted as an interoceptive biomarker, yet its two best-documented physiological modulators, sleep depth and age, are uncharacterised topographically and unquantified at scale, and their behaviour in disease is unknown. Without that baseline, HEP differences reported between patient and control groups cannot be attributed to altered cortical interoception rather than to unmatched age, unbalanced sleep-stage composition, or the cardiac field artifact. This proposal establishes the normative sleep-stage and age topography of the HEP in a large corpus and then tests, within the same framework, whether and how clinical diagnosis changes it.

2. Research objectives and hypotheses

Central question. Do the sleep-stage and age gradients of the heartbeat-evoked potential reflect a single, topographically organised axis of cortical interoceptive gain, and is that axis preserved or altered in patients with cardiovascular, cerebrovascular and neurodegenerative disease?

From this, four sub-questions:

- **Q1, Sleep-stage topography.** Across which electrodes, and in which post-R-peak time window, does the HEP differ between REM, light sleep and deep sleep? Hypothesis (H1): REM and light sleep differ minimally, while both diverge sharply from deep sleep over fronto-central sites, i.e. the HEP tracks arousability rather than a graded sleep-depth continuum.
- **Q2, Age.** Within a single sleep stage (REM), does an older-vs-younger contrast produce a spatial pattern that overlaps the deep-sleep–REM pattern of Q1? Hypothesis (H2): the older-minus-younger REM difference is spatially correlated with, and comparable in magnitude to, the deep-sleep–REM difference, ageing shifts interoceptive gain along the same axis as sleep depth.
- **Q3, Shared generator.** Do the Q1 and Q2 effects localise to a common cortical source? Hypothesis (H3): both project to insular / somatosensory-opercular and cingulate sources, consistent with a single central-interoceptive network being up- and down-regulated.
- **Q4, Diagnosis.** For each diagnosis group (atrial fibrillation, heart failure, stroke, cognitive impairment / dementia), matched to a diagnosis-free cohort for age and sex, is the sleep-stage HEP profile (a) unchanged, (b) uniformly rescaled, or (c) reshaped (a different topography or time course)? Hypothesis (H4): most diagnoses rescale the normative gradient without reshaping it (the delta waveform stays collinear with the reference), whereas conditions with direct central-autonomic involvement, stroke and dementia, show a genuine change in topography.

The objectives are ordered so that a negative result at one stage still leaves the later stages answerable: Q1–Q2 are descriptive and will yield an interpretable map regardless of outcome; Q3 depends on source-localisation feasibility (addressed in Risks); Q4 is the translational payoff and is designed to be robust to the exact form of the Q1 result.

3. Approaches, stages of implementation

The work is organised into five stages that map onto the four research questions plus a consolidation stage. Each stage below states its rationale, the key analytic choices, the expected duration, the foreseen difficulties, and the fallback if the primary plan fails. Durations assume a four-year Ph.D. and are summarised in the Gantt chart in Section 6.

Stage 0, Corpus assembly and pipeline freeze (months 1–6)

Rationale: every downstream contrast must run on identically processed data, so the pipeline is fixed and version-locked before any hypothesis test. Tasks: ingest EDF recordings and hypnograms; identify EEG and ECG channels by a 10–20 / 10–10 name whitelist (excluding intracranial and auxiliary channels present in this heterogeneous corpus); band-pass filter (0.5–100 Hz) and notch at the detected line frequency with MNE-Python (Gramfort et al., 2013); detect R-peaks; epoch –300 to +400 ms around each R-peak; reject bad epochs with AutoReject; standardise per channel. Link diagnosis codes and demographics from the hospital records. Deliverable: a frozen per-patient, per-sleep-stage HEP cache with a documented quality-control manifest. Difficulty: channel-naming and montage heterogeneity across sites; mitigated by the whitelist plus manual audit of a random 200-recording sample. Fallback: if full-montage yield is lower than expected, restrict topographic analyses to the six consistently present sites (F3/F4/C3/C4/O1/O2) and treat the full montage as a secondary analysis.

Stage 1, Sleep-stage HEP topography, Q1 (months 6–14)

Within each patient, compute the per-stage HEP evoked average per electrode, then the paired REM–deep-sleep, REM–light-sleep and light-sleep–deep-sleep differences. Test each with a cluster-mass permutation test over electrodes $\times$ time (Maris and Oostenveld, 2007), 200–000–500 permutations, $\alpha = 0.01$, with the ± 50 ms QRS window excluded from every test. Report per-electrode corrected p-value topomaps and the peak significant latency. Difficulty: unequal epoch counts per stage bias the evoked-average noise floor; mitigated by epoch-count matching within patient before averaging and by a non-heartbeat-locked surrogate-event control that re-epochs the same window around random pseudo-events. Fallback: if paired within-patient data are too sparse for one stage pair, fall back to an age-and-sex-matched between-patient contrast.

Stage 2, Age contrast and axis overlap, Q2 (months 12–20)

Within REM only, split patients at the median age and run the same electrode-wise permutation contrast (older – younger). Quantify the spatial overlap between the Stage-2 map and the Stage-1 REM–deep-sleep map by the Pearson correlation of their per-electrode effect sizes, with a spin/permutation null. Repeat with age as a continuous predictor in a per-electrode linear mixed model (fixed effects: age, sex, stage; random intercept: patient;

random slope: site) to check that the median split is not creating the effect. Difficulty: age confounds with diagnosis burden and medication; mitigated by repeating the contrast within the diagnosis-free subset. Fallback: if the continuous model does not converge, report the tertile ANOVA used in preliminary work.

Stage 3, Source localisation, Q3 (months 20–28)

Project the significant sensor-level differences to cortex with a template-MRI boundary-element forward model and dSPM/eLORETA (Gramfort et al., 2013), and test whether the Stage-1 and Stage-2 source maps overlap in insular, opercular-somatosensory and cingulate regions of interest. Difficulty: this is a clinical corpus without individual MRIs and with a sparse montage, so source estimates are coarse; this stage is explicitly exploratory. Fallback: if source localisation is judged unreliable, substitute a sensor-space generator analysis, spatial-pattern similarity, topographic PCA, and comparison against published HEP source topographies, which is sufficient to address whether the two effects share a spatial signature.

Stage 4, Diagnosis comparison, Q4 (months 24–42)

For each of the four index diagnoses, build a diagnosis-free comparison cohort matched on age (± 3 years), sex and recording site by nearest-neighbour matching. Three complementary analyses per diagnosis: (i) the Stage-1 electrode-wise permutation contrast, run diagnosis vs matched reference within each sleep stage; (ii) a waveform-alignment permutation test that asks whether the diagnosis group's sleep-stage delta waveform is collinear with the reference delta (observed cosine / Pearson distance vs a label-shuffling null), a positive result means 'rescaled, not reshaped'; (iii) a cross-validated classifier (logistic regression and gradient boosting) trained on HEP + heart-rate + T-wave features to test whether diagnosis is decodable from the interoceptive profile, with permutation-based chance estimation and feature-importance reporting. Difficulty: comorbidity overlap (many patients carry several index diagnoses); mitigated by a 'single-diagnosis-only' sensitivity analysis and by modelling comorbidity count as a covariate. Fallback: if matched-cohort sizes are too small for a given diagnosis, pool into broader categories (any-cardiac, any-cerebrovascular, any-neurodegenerative).

Stage 5, Consolidation, CFA control analyses, and writing (months 40–48)

Re-run every headline contrast with (a) an ECG-informed ICA cleaning step and (b) a per-channel regression of the HEP evoked average on the patient's own ECG evoked average, confirming that each reported effect survives both and peaks outside the QRS window. Assemble three manuscripts (sleep-stage/age topography; source/generator; diagnosis comparison) and the dissertation.

4. Materials and methods

Data. The Human Sleep Project multi-centre clinical polysomnography corpus (~90,166 overnight recordings; ~2,443 with the full 19-electrode 10–20 montage), accessed under approval from the Brain Data Science Platform (bdsp.io/content/hsp). Each recording provides synchronous scalp EEG and ECG, a technician-scored hypnogram, and linked demographic and ICD diagnosis codes. No new data are collected.

Design. Retrospective, observational, within- and between-subject contrasts on existing recordings; no intervention. Primary units of analysis are per-patient, per-sleep-stage, per-electrode HEP evoked averages.

Signal processing. EEG/ECG read and filtered (0.5–100 Hz band-pass, harmonic line notch) with MNE-Python (Gramfort et al., 2013); resampling to 256 Hz; bad-channel repair via the PREP pipeline (Bigdely-Shamlo et al., 2015) then interpolation; residual-artifact removal with AutoReject (Jas et al., 2017); per-channel standardisation. HEP epochs –300 to +400 ms about each R-peak; ± 50 ms QRS window excluded from all tests (Steinfath et al., 2026).

Statistics. Nonparametric cluster-mass permutation tests over electrodes $\times$ time (Maris and Oostenveld, 2007), $\alpha = 0.01$; per-electrode linear mixed models for continuous age; Mann–Whitney / Kruskal–Wallis for skewed scalar summaries with Benjamini–Hochberg FDR control; nearest-neighbour cohort matching for diagnosis contrasts; cross-validated logistic regression and gradient boosting (scikit-learn; Pedregosa et al., 2011) with permutation chance estimation for the decoding analysis.

Computing. Analyses run on the laboratory's Linux compute server; all pipeline code is version-controlled and re-used unchanged across contrasts.

5. Preliminary results

The following results are from the student's own unpublished analyses of the corpus and establish that the experimental system can answer the research questions. They are summarised in Figure 1 at the end of this document.

5.1 A robust, wide-field sleep-stage HEP difference

In $N = 2,136$ patients with paired REM and deep-sleep epochs, the REM–deep-sleep HEP difference was significant by cluster-mass permutation test ($p = 0.005$; significant at 15/19 electrodes), with the effect peaking well outside the excluded QRS window. The REM–light-sleep difference was markedly smaller and did not survive threshold ($N = 2,202$; $p = 0.015$; 1/19 electrodes). This two-tier pattern, REM $\approx$ light sleep $\blacksquare$ deep sleep, is the basis for hypothesis H1 and shows the corpus has the power and montage coverage for electrode-wise topographic inference.

5.2 An age effect of comparable magnitude, along the same axis

Within REM, an older-vs-younger median-split contrast ($N = 1,144$ vs $1,299$) was significant ($p = 0.005$; 9/19 electrodes) and comparable in magnitude to the REM–deep-sleep contrast of 5.1. This is the preliminary support for H2: ageing appears to move the HEP along the same sleep-depth axis rather than producing an orthogonal pattern. Formal spatial-overlap testing (Stage 2) has not yet been run.

5.3 First diagnosis comparison: rescaling, not reshaping

An initial diagnosis analysis matched a diagnosis-free cohort ($n = 2,634$) to four diagnosis subgroups, atrial fibrillation ($n = 1,240$), heart failure ($n = 1,658$), stroke / cerebrovascular ($n = 1,701$) and cognitive impairment / dementia ($n = 947$), and compared the sleep-stage HEP delta vectors. For atrial fibrillation the diagnosis delta was almost perfectly collinear with the diagnosis-free delta (Pearson $r = 0.9995$; cosine similarity 0.9977; alignment permutation $p \approx 0.94$, i.e. no detectable change in shape). This directly motivates Q4's three-way ‘unchanged / rescaled / reshaped’ framing and suggests that the interesting deviations, if any, will be concentrated in the central-autonomic diagnoses (stroke, dementia), which is where Stage 4 is powered to look.

6. Timeline and tasks

Durations are in project months over a four-year Ph.D. Overlap between stages is intentional: analysis of one stage proceeds while the next stage's data are prepared.

Stage / task	Objective	Months	Y1	Y2	Y3	Y4
0 Corpus assembly & pipeline freeze	Infrastructure	1–6	■			
1 Sleep-stage HEP topography	Q1 / H1	6–14	■	■		
2 Age contrast & axis overlap	Q2 / H2	12–20		■		
3 Source / generator analysis	Q3 / H3	20–28		■	■	
4 Diagnosis comparison	Q4 / H4	24–42			■	■
5 CFA controls, consolidation, writing	All	40–48				■
Manuscript 1 (topography)		16–22		■		
Manuscript 2 (generator)		28–34			■	
Manuscript 3 (diagnosis) + thesis		42–48				■

7. Risks and solutions

- **Cardiac field artifact drives a group effect.** Every headline contrast is re-run with ECG-informed ICA and with per-channel HEP-on-ECG regression, and only effects that survive both and peak outside the ± 50 ms QRS window are reported. Diagnosis contrasts are additionally age/sex/site matched, since CFA covaries with body habitus and sex.
- **Cohort heterogeneity / montage sparsity.** If full-19-electrode yield is low, topographic claims are restricted to the six consistently covered sites and the full montage becomes a secondary analysis; between-site effects are modelled as a random factor.
- **Diagnosis subgroups too small after matching.** Pool into broad categories (any-cardiac, any-cerebrovascular, any-neurodegenerative); report the matched sample size and minimum detectable effect for every contrast.
- **Comorbidity confounding.** Sensitivity analysis on single-diagnosis-only patients plus comorbidity-count covariate; pre-register the primary contrast for each diagnosis.

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Supporting figures

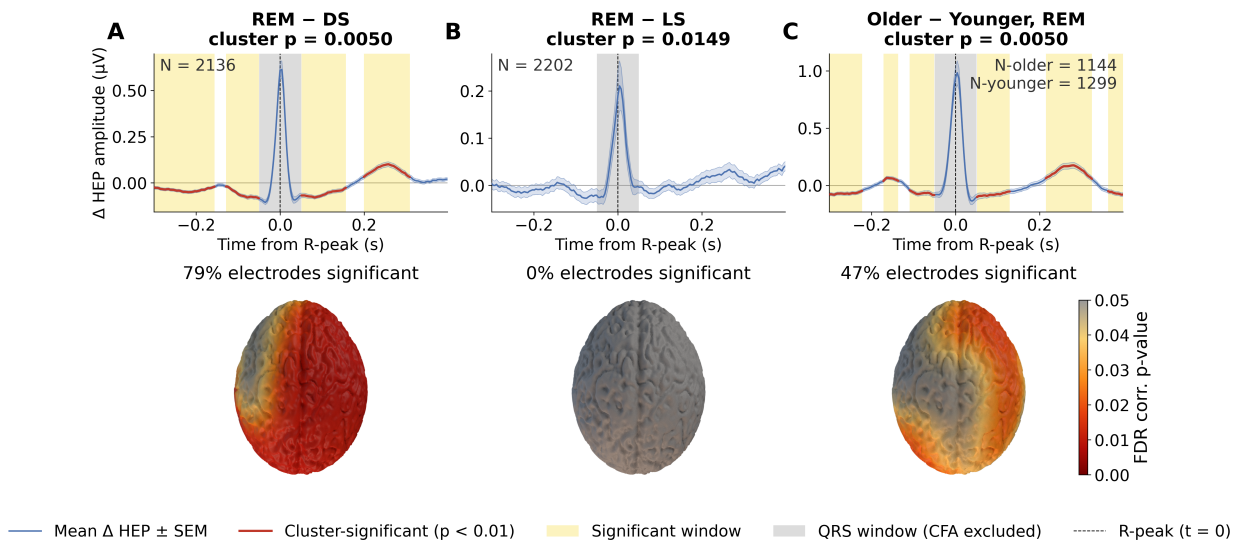


Figure 1. Preliminary sleep-stage and age HEP contrasts (student's unpublished analysis). Each panel: mean Δ HEP waveform (left) and per-electrode corrected-significance topomap (right); red = cluster-significant ($p < 0.01$), grey = QRS window excluded from testing. A: REM – deep sleep ($N = 2,136$; cluster $p = 0.005$; 15/19 electrodes). B: REM – light sleep ($N = 2,202$; $p = 0.015$; 1/19). C: older – younger within REM ($N = 1,144$ vs $1,299$; $p = 0.005$; 9/19). All effects peak outside the excluded QRS window.